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Activity # 1 - Variables

PART A- VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	Stored In
s1	student	Reference Variable	stack
s2	student	Reference Variable	stack
s3	student	Reference Variable	stack
name	string	Instance Variable	Heap
grade	int	Instance Variable	Heap
n	string	Constructor Parameter	stack
g	int	Constructor Parameter	stack

PART B- EXPLANATION

Statement	What Happens?	Memory Impact
• student s1 = new student ("John", 85);	New student object created with name "John" and grade = 85. s1 then references the object.	New object in Heap. s1 stored in stack.
• student s2 = new student ("Maria", 90);	Another student object created with name "Maria" and grade = 90. s2 then references the object.	New object in Heap. s2 stored in stack.
• student s3 = s2;	s3 will reference the same object as s2.	s3 stored in stack pointed to s2's object.
• s1.grade = 95;	Updates the grade of the object referenced by s1. 85 → 95	Modifies existing Heap object.
• s3.name = "Ana"	Changes the name of the object referenced by s3 → s2. Maria becomes Ana.	same Heap object updated; affects both s2 and s3.

PART C- OUTPUT

Code output:

- s1 → ("John", 95)
- s2 → ("Ana", 90)
- s3 → ("Ana", 90)

→

John 95

Ana 90

Ana 90